



Positive existential definability in finite structures

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ABSTRACT

For a finite structure M on the domain of $k \geq 2$ elements consider the description of the Galois closures $\text{InvEnd}(M)$ and $\text{InvAut}(M)$ (the set of relations on the domain of M that are invariant to all endomorphisms, resp. automorphisms of M) via positive existential formulae over the signature of M . It is shown that for every relation R from the set $\text{InvEnd}(M)$ (R from $\text{InvAut}(M)$) there exists a positive existential formula over M (resp., over M and the inequality predicate) with no more than $(k-1)m$ (m is the size of R) existential quantifiers that defines R . This implies that every first order n -ary ($n \geq 1$) predicate over M (i.e., with n free variables) is equivalent to a positive existential formula over M and inequality with at most $(k-1)(k^n-1)$ existential quantifiers and this estimate does not depend on the signature of M . Next, for a finite M with a finite signature there is a cubic algorithm that for every R on the domain of M it recognizes properties: R belongs to $\text{InvAut}(M)$ (that is, R is defined by some first order predicate over M), as well as R belongs to $\text{InvEnd}(M)$. Moreover, for such structure M there is a quadratic algorithm that for every R from $\text{InvAut}(M)$ (R from $\text{InvEnd}(M)$) it produces a positive existential formula over the unnested atomic predicates from M and the inequality $x \neq y$ (resp., over unnested atomic predicates from M), which defines R in $mO(n^2)$ steps, where n is the arity of R .

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1. Introduction

Let M be a finite structure on the domain E of $k \geq 2$ elements. Consider three types of Galois closures on a set of all relations defined on the same finite domain:

- (1) $\text{InvPol}(M)$: all relations on E , which are invariant to all *polymorphisms* of the set of all atomic relations from M (denote this set also by M). Relations from $\text{InvPol}(M)$ are expressed by all *primitive positive* (p.p.) formulae over the signature of M (see [2] and [7, (4.3)]), i.e., $(\exists, \wedge, =)$ -formulas, also called *conjunctive queries* in the database theory.
- (2) $\text{InvEnd}(M)$: all relations on E , which are invariant to all *endomorphisms* of M . Such relations are expressed by disjunction of all p.p. formulae over M (see [2]) that is equivalent to *positive existential* formulae over M (i.e., formulae built by operations $\wedge, \vee$, and $=$, which have a string of existential quantifiers in front of them (see [4]).
- (3) $\text{InvAut}(M)$: all relations on E , which are invariant to all *automorphisms* of M . For a finite structure such relations are exactly those which are defined by the first order formulae over M (see [2,7]). Moreover, the set of these relations on the finite domain coincides (see [7, Proposition 4.2]) with the set of all relations defined by *existential* [4] formulae over M , that is, quantifier-free formulae with a string of existential quantifiers in front of them.

Then for a finite structure with a finite signature (having a finite number of base relations, operations, and constants) we consider *decidability* and *solvability* problems related to the above closures (1), (2), and (3). Namely, does there exist an

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algorithm that for a finite structure M with a finite signature and an n -ary ($n \geq 1$) relation R on the domain of M (from the viewpoint of applications we may consider R as a set of observations, or a state of knowledge) such that:

- (a) it verifies properties $R \in \text{InvPol}(M)$, $R \in \text{InvEnd}(M)$, and $R \in \text{InvAut}(M)$ (decidability problem);
- (b) if it is true for some of the above property, then builds a formula in the corresponding logical formalism (see cases (1), (2), and (3)) that defines R (solubility problem)?

All these problems are decidable and solvable for three Galois closures. Moreover, it is shown in this paper, that for the closures $\text{InvEnd}(M)$ and $\text{InvAut}(M)$ there is a cubic algorithm on their decidability, and a quadratic algorithm on their solubility, in the case, when the input of that algorithm is an n -ary ($n \geq 1$) relation R , not a structure M itself.

The review on the complexity of different types of decidability problems in this field can be found in [3]. The complexity of a p.p. definability decision problem for finite relational structures was studied in [8] (it is hard for non-Boolean domains).

Furthermore, in a general case of any signature of M (finite or infinite) we show that for every relation $R \in \text{InvEnd}(M)$ ($R \in \text{InvAut}(M)$) there exists a positive existential formula over M (resp., over M and the inequality predicate $x \neq y$) with no more than a string of $(k-1)|R|$ ($|E| = k \geq 2$, $|R|$ is a size of R) existential quantifiers that defines R . It implies that every first order n -ary ($n \geq 1$) predicate constructed over M is equivalent in M to a positive existential formula over M and $x \neq y$, having at most $(k-1)(k^n-1)$ existential quantifiers, which does not depend on the signature of M , but only on the size of its domain.

Note that we avoid using the universal quantifier $\forall$ in our predicate formalism, since the existential quantifier $\exists$ is more frequently utilized in Logic applications and it is much easier to implement in algorithmic setting. Hence, we consider three types of formulae: $(\exists, \wedge, =)$ -formula that stands for a p.p. formula, $(\exists, \wedge, \vee, =)$ -formula that stands for a positive existential formula, and $(\exists, \wedge, \vee, \neg, =)$ -formula that stands for the first order predicate formula.

2. Galois closures on finite structures

Let $M = (E, \mathfrak{R}, \mathfrak{J}, \mathbf{C})$ be a finite structure on the domain E , $|E| = k \geq 2$, where $\mathfrak{R}$ is a set of relations on E , $\mathfrak{J}$ is a set of everywhere defined operations on E , $\mathbf{C}$ is a set of constants (0-ary operations). In what follows, we denote by the same symbol M the set of all relations expressed by atomic predicates over M , excluding atomic sentences, that is, the set of all non-empty relations on E of arity $m \geq 1$ defined by predicates of the form $R(F_1(x_1, \dots, x_m), \dots, F_n(x_1, \dots, x_m))$ or $F_1(x_1, \dots, x_m) = F_2(x_1, \dots, x_m)$, where $R \in \mathfrak{R}$ and $F_1, \dots, F_n$ ($n \geq 1$) are m -ary terms over M . These terms are represented by all operations from a clone $[\mathfrak{J} \cup \mathbf{C}]$ on E generated via composition with projections from the set $\mathfrak{J}$ with application of constants from $\mathbf{C}$ (that is, substituting elements named by constants from $\mathbf{C}$ in place of variables in operations from the clone $[\mathfrak{J}]$ generated by $\mathfrak{J}$). To keep a uniform approach, for a structure $M = (E, \mathfrak{R}, \mathbf{C})$ (a relational structure with constants) we consider $[\mathbf{C}]$ as a trivial clone of projections with application of constants to them, e.g., a binary atomic formula $R(\mathbf{c}, x_1, x_2)$ ($\mathbf{c} \in \mathbf{C}$) is equivalent to (defines the same atomic relation) $R(e_3^1(c, x_1, x_2), e_2^1(x_1, x_2), e_2^2(x_1, x_2))$, where $\mathbf{c}$ names an element $c \in E$, as well as the equality predicate $x_1 = x_2$ is equivalent to $e_2^1(x_1, x_2) = e_2^2(x_1, x_2)$, where $e_n^j(x_1, \dots, x_n) = x_j$ ($1 \leq j \leq n$) is an n -ary ($n \geq 2$) projection operation on E .

An m -ary ($m \geq 1$) everywhere defined operation f on E preserves an n -ary ($n \geq 1$) relation R on E if

$$\begin{aligned} R(a_{11}, \dots, a_{1n}) \wedge \dots \wedge R(a_{1m}, \dots, a_{mn}) \wedge f(a_{11}, \dots, a_m) = b_1 \wedge \dots \\ \wedge f(a_{n1}, \dots, a_{nm}) = b_n \rightarrow R(b_1, \dots, b_n) \end{aligned} \quad (2.1)$$

holds for all $a_{ij} \in E$, $b_j \in E$ ($1 \leq i \leq m$, $1 \leq j \leq n$).

We call that f a *polymorphism* (see [2]) of R , denote it by $f \in \text{Pol}(R)$. If f preserves all $R \in M$, then f is a polymorphism of a structure M , $f \in \text{Pol}(M)$. In case, when f is a unary operation preserving R (M), then f is an *endomorphism* of R , $f \in \text{End}(R)$ (resp., of M , $f \in \text{End}(M)$). If f is a unary bijective operation, then f is an *automorphism* of R , $f \in \text{Aut}(R)$ (resp., of M , $f \in \text{Aut}(M)$). Note that in the case of a finite domain the property “a bijective f preserves R ” is equivalent to a stronger definition of an automorphism: $R(a_1, \dots, a_m) \iff R(fa_1, \dots, fa_m)$ holds for all $a_i \in E$ ($1 \leq i \leq n$) (e.g., see [7]). This holds, since each power of f , including f^{-1} on the finite domain, also preserves R , that is equivalent to the last condition. Denote $L_{\omega\omega}(M)$ ($L_{\omega 0}(M)$) the set of all relations defined by the first order (resp., quantifier-free) formulae from atomic relations of M . Then $L_{\omega 0}^{\exists}(M)$ is a set of all existential relations built from M . Then we have the following properties (see [2,7]).

Proposition 2.1. *Let M be a finite structure. Then*

- (1) $\text{InvAut}(M) = L_{\omega\omega}(M)$.
- (2) $L_{\omega\omega}(M) = L_{\omega 0}^{\exists}(M)$.
- (3) $R \in \text{InvPol}(M) \iff R$ is defined by a p.p. formula over M .
- (4) $R \in \text{InvEnd}(M) \iff R$ is defined by a positive existential formula over M .

In general, a finite structure M with functional symbols in its signature may have an infinite (countable) number of atomic relations which are not equivalent up to fictitious (full) or equal coordinates; that creates some difficulties in dealing, with regard to algorithmic complexity, with the sets $\text{Pol}(M)$, $\text{End}(M)$, and $\text{Aut}(M)$. So, we simplify the definitions of those sets in the following way. Let $\mathcal{R}$ be a finite set of all *unnested* (see [4]) predicates over M :

$$\mathcal{R} = \{\mathfrak{R}, x = \mathbf{c}(\mathbf{c} \in \mathbf{C}), h(x_1, \dots, x_l) = x_{l+1}(h \in \mathfrak{J})\}.$$

Denote $\mathcal{R}_M$ a set of all unnested atomic relations defined by predicates from $\mathcal{R}$. Then from [7, Corollary 4.6] we obtain $\text{Pol}(M) = \text{Pol}(\mathcal{R}_M)$, $\text{End}(M) = \text{End}(\mathcal{R}_M)$, and, consequently, $\text{Aut}(M) = \text{Aut}(\mathcal{R}_M)$. Next, we utilize two properties of the above Galois closures (see [7]): $\text{End}(\mathcal{R}_M) = \text{Pol}(\{\mathcal{R}_M \cup \{x = y \vee x = z\}\})$ (up to fictitious variables in operations) and $\text{Aut}(\mathcal{R}_M) = \text{Pol}(\{\mathcal{R}_M \cup \{x = y \vee x = z\} \cup \{x \neq y\}\})$ (up to fictitious variables in operations), where $x = y \vee x = z$ and $x \neq y$ stand for a ternary and a binary relation defined by corresponding predicates. Hence, we summarize the above results.

Proposition 2.2. ([2,7]). *Let M be a finite structure and R be a non-void relation on the same domain. Then*

- (1) $R \in \text{InvPol}(M) \Leftrightarrow R$ is defined by a p.p. formula from $\mathcal{R}_M$.
- (2) $R \in \text{InvEnd}(M) \Leftrightarrow R$ is defined by a p.p. formula from $\mathcal{R}_M \cup \{x = y \vee x = z\}$.
- (3) $R \in \text{InvAut}(M) \Leftrightarrow R$ is defined by a p.p. formula from $\mathcal{R}_M \cup \{x = y \vee x = z\} \cup \{x \neq y\}$.
- (4) all the above formulas from (1)–(3) defining R without repetition of coordinates have the string of existential quantifiers of the length $k^{|R|} - n$, where $|E| = k \geq 2$, $1 \leq |R| \leq k^n$, and $n \geq 1$ is the arity of R .

Proposition 2.3. ([2,7]). *Let M be a finite structure with a finite signature. Then*

- (a) *The properties $R \in \text{InvPol}(M)$, $R \in \text{InvEnd}(M)$, and $R \in \text{InvAut}(M)$ are decidable.*
- (b) *There exists an algorithm such that for every above property presents a corresponding p.p. formula with the prefix of $k^{|R|} - n$ existential quantifiers for a non-full relation R without repetition of coordinates.*

For the decidability of $\text{InvPol}(M)$ we use a chain, based on the Local theorem (see [5]): $R \in \text{InvPol}(\mathcal{R}_M) \Leftrightarrow \text{Pol}(\mathcal{R}_M) \subseteq \text{Pol}(R) \Leftrightarrow \text{Pol}^{(n)}(\mathcal{R}_M) \subseteq \text{Pol}^{(n)}(R)$ ($|R| = n \geq 1$), where $\text{Pol}^{(n)}(\mathcal{R}_M)$ ($\text{Pol}^{(n)}(R)$) are all n -ary operations preserving $\mathcal{R}_M$ (resp., R), that is, every n -ary polymorphism of $\mathcal{R}_M$ is a polymorphism of R . The decidability of $\text{InvEnd}(M)$ and $\text{InvAut}(M)$ follows straight from $\text{End}(M) = \text{End}(\mathcal{R}_M)$ and $\text{Aut}(M) = \text{Aut}(\mathcal{R}_M)$, since $\mathcal{R}_M$ is a finite set of relations and, therefore, we may find all elements from $\text{End}(M)$ and $\text{Aut}(M)$ in a finite number of steps. Then we use the following property: $R \in L_{\omega\omega}(M)$ if and only if $\text{Aut}(M) \subseteq \text{Aut}(R)$. In the next Section we show that the decidability problems for $\text{InvEnd}(M)$ and $\text{InvAut}(M)$ can be solved in $O(n^2|R|^3)$ time. In Boolean case ($k = 2$), the decidability of $\text{InvPol}(\mathfrak{A})$, $\mathfrak{A}$ is a finite set of relations on the Boolean domain, can be solved in quadratic time $O(n^2|R|^2)$ (see [3]).

Note that $k^{|R|} - n > 0$ for every non-full relation R without repetition of coordinates. The case of repetition of coordinates and full coordinates in the algorithm from Proposition 2.3(b) can be handled by a conjunction with the equality relation without additional quantifiers. At the same time, this number is too high to construct an effective algorithm depending on parameters n and $|R|$. So, in the next Section we apply a modification of the algorithm from [2] in case of endomorphisms (automorphisms) of M with a finite signature and we obtain a quadratic algorithm with an upper estimate $(k-1)|R|$ of the prefix of existential quantifiers in a positive existential formula (resp., existential formula) defining R , which does not depend on the signature of M .

2.1. Partial structures

Consider an extension of the notion of a structure called a *partial structure* (see [6]) $M' = (E, \mathfrak{A}, \mathfrak{I}', \mathbf{C})$, where $\mathfrak{I}'$ is a set of partial operations on E , that includes both everywhere defined, as well as not everywhere defined operations. In that case, some terms over M' present not everywhere defined operations on E , which affects the definition of atomic predicates (see [6]): if any n -ary ($n \geq 1$) term involved in the construction of an atomic predicate is not defined on some n -tuple on E , then this predicate is false on this n -tuple (includes the case when a closed term may result in an empty operation).

Then similar to the definition of $\mathcal{R}_M$ we consider the set $\mathcal{R}_{M'}$ of all unnested atomic relations from M' (see [6]) and applying the proof from [7, Corollary 4.6] we obtain $\text{Pol}(M') = \text{Pol}(\mathcal{R}_{M'})$, $\text{End}(M') = \text{End}(\mathcal{R}_{M'})$, and $\text{Aut}(M') = \text{Aut}(\mathcal{R}_{M'})$. Hence, following the proofs of the paper we may obtain that the main results and algorithms from it are also valid in the case of finite partial structures (just substitute $\mathcal{R}_M$ by $\mathcal{R}_{M'}$).

3. Positive existential representation

For Galois closures $\text{InvEnd}(M)$ and $\text{InvAut}(M)$ for a finite M with a finite signature we may strengthen Propositions 2.2 and 2.3.

Theorem 3.1. *Let M be a finite structure of cardinality $k \geq 2$ having a finite signature. Then there exists a quadratic algorithm such that for every n -ary ($n \geq 1$) $R \in \text{InvAut}(M)$, by using $|R| \cdot O(n^2)$ steps, it presents a positive existential formula over the set $\mathcal{R}$ of all unnested atomic predicates from M and $x \neq y$, which defines R . Moreover, such formula contains no more than $(k-1) \cdot |R|$ existential quantifiers.*

Proof. Without loss of generality, we put $E = \{0, 1, \dots, k-1\}$, $k \geq 2$. Then the main k -orbit of the group $\text{Aut}(M)$ of permutations on E is a k -ary relation $Gp(M) = \{(a_1, \dots, a_k) \mid (a_1, \dots, a_k) = (\varphi(0), \dots, \varphi(k-1)) \text{ for all } \varphi \in \text{Aut}(M)\}$. The proof of the Theorem is based on the two following Lemmas.

Lemma 3.2. Let M be a finite structure with a finite signature. Then there exists an algorithm, that is creating a k -ary $(\wedge, =)$ -formula $Gp(x_0, \dots, x_{k-1})$ over the set $\mathcal{R}$ of all unnested atomic predicates from M and $x \neq y$, which defines the main k -orbit $Gp(M)$ of $\text{Aut}(M)$.

Proof of the Lemma. From [7, Corollary 4.6] we have:

$$\text{InvPol}(M) = \text{InvPol}(\mathcal{R}_M).$$

Hence, for the 1-graph of $\text{Pol}(M)$ (see [2,7]), that is, the k -ary relation $G_1(\text{Pol}(M)) = \{ (a_1, \dots, a_k) \mid (a_1, \dots, a_k) = (\psi 0, \dots, \psi(k-1)) \text{ for all } \psi \in \text{End}(M) \}$ defined by the monoid $\text{End}(M)$ of all endomorphisms of M , we have $G_1(\text{Pol}(M)) = G_1(\text{Pol}(\mathcal{R}_M))$. Then we construct $G_1(\text{Pol}(\mathcal{R}_M))$ utilizing the procedure from [2] over the set of relations $\mathcal{R}_M$.

We form a Cartesian (direct) product $\bar{R} = R_1 \otimes \dots \otimes R_l$ of all unnested atomic relations from $\mathcal{R}_M = \{R_1, \dots, R_l\}$, which is expressed by a repetition-free conjunction of all unnested atomic predicates from M . Denote by $s = |R_1| \times \dots \times |R_l|$ the number of all n -tuples ($n \geq 1$) of the n -ary relation $\bar{R}$, where $n = \text{arity } R_1 + \dots + \text{arity } R_l$. Then $\bar{R} = \{r_1, \dots, r_s\}$ consists of s n -tuples r_i ($i = 1, \dots, s$) over E . Clearly, $\text{End}(\mathcal{R}_M) = \text{End}(\bar{R})$.

Next, from $\bar{R}$ we form such repetition-free $\wedge$ -formula with ns variables:

$$R^* \equiv \bar{R}(x_0, \dots, x_{n-1}) \wedge \bar{R}(x_n, \dots, x_{2n-1}) \wedge \dots \wedge \bar{R}(x_{(n-1)s}, \dots, x_{ns-1}). \quad (3.1)$$

Clearly, $\text{End}(\bar{R}) = \text{End}(R^*)$. Then from (3.1) we have $\bar{r} = r_1 \otimes \dots \otimes r_s \in R^*$, where $\bar{r}$ is a Cartesian product of all n -tuples from $\bar{R}$. Next, we implement an identification (mapping) $\Delta: \{x_0, \dots, x_{ns-1}\} \rightarrow \{x_0, \dots, x_{k-1}\}$ of ns variables from R^* into its subset of k variables, namely, $(\Delta R^*)(x_0, \dots, x_{k-1}) \equiv R^*(\Delta(x_0), \dots, \Delta(x_{ns-1}))$ (it is a part of a $\wedge$ -formula) such that $\Delta(x_i) = x_j \leftrightarrow r^i = j$ ($\bar{r} = (r^0, \dots, r^{ns-1})$, $0 \leq i \leq ns-1$, $0 \leq j \leq k-1$), which, in case when Δ is onto mapping, defines a k -ary relation ΔR^* with the property $(0, 1, \dots, k-1) \in \Delta R^*$. In case when there exists m ($0 \leq m \leq k-1$) that $r^i \neq m$ for all $i = 0, \dots, ns-1$ (x_m does not belong to the range of Δ), we perform, in addition, such conjunction $\Delta R^* \wedge x_m = x_m$, i.e., add the full m th coordinate to ΔR^* .

From an algorithmic standpoint, it is helpful to describe the procedure of construction of ΔR^* in terms of matrices of relations, provided R^* is presented as an $s^n \times ns$ matrix $[R^*]$ over E with s^n rows that are all ns -tuples from R^* (the order in which we put rows is not essential). Then, using the tuple $\bar{r}$, we introduce an ns -ary diagonal relation defined by this $\wedge$ -formula: $D(\bar{r})(x_0, \dots, x_{ns-1}) \equiv \bigwedge (x_i = x_j) (r^i = r^j: 0 \leq i < j \leq ns-1)$ and cross-out in $[R^*]$ all rows (ns -tuples) that do not belong to the matrix $[D(\bar{r})]$, i.e., obtain the matrix of the intersection $R^* \cap D(\bar{r})$ of ns -ary relations R^* and $D(\bar{r})$. Next, we cross-out in the matrix $[R^* \cap D(\bar{r})]$ all duplicate columns and rearrange the k remaining columns (permutation of coordinates) such that the k -tuple $(0, 1, \dots, k-1)$ belongs to the resulting relation ΔR^* .

Finally, we prove the equality of k -ary relations:

$$\Delta R^* = G_1(\text{Pol}(M)) \quad (3.2)$$

Proof of (3.2). Indeed, it is easy to verify that the ns -ary relation $R^* \cap D(\bar{r})$ consists of all direct products $r_{i_1} \otimes \dots \otimes r_{i_s}$ ($r_{i_1}, \dots, r_{i_s} \in \bar{R}$) of s not necessarily distinct n -tuples from $\bar{R}$ such that any equal pair of coordinates in $r_1 \otimes \dots \otimes r_s$ corresponds to an equal pair of coordinates in $r_{i_1} \otimes \dots \otimes r_{i_s}$. Hence, for each product $r_{i_1} \otimes \dots \otimes r_{i_s} \in \bar{R}$ there exists a unary mapping $\psi: E \rightarrow E$ such that $r_{i_1} = \psi(r_1), \dots, r_{i_s} = \psi(r_s)$. It is equivalent to the property of ψ to be an endomorphism of the relation $\bar{R}$: $\psi(r_i) \in \bar{R}$ for all $r_i \in \bar{R}$ ($1 \leq i \leq s$) (i.e., $\psi \in \text{End}(\mathcal{R}_M) = \text{End}(M)$), and, consequently, to be an endomorphism of $R^* \cap D(\bar{r})$, as well as an endomorphism of ΔR^* . Since the k -ary ΔR^* is obtained from ns -ary $R^* \cap D(\bar{r})$ by identification of duplicate coordinates (in particular, the k -tuple $(0, 1, \dots, k-1) \in \Delta R^*$ is transformed by the above operation from the product $r_1 \otimes \dots \otimes r_s \in R^*$) we have $\text{End}(R^* \cap D(\bar{r})) = \text{End}(\Delta R^*)$, and, consequently, we obtain that each unary mapping ψ such that $\psi(0, 1, \dots, k-1) \in \Delta R^*$ belongs to $\text{End}(M)$, and vice versa for each $\psi \in \text{End}(M)$ we have $\psi(0, 1, \dots, k-1) \in \Delta R^*$. $\square$

Then from the k -ary $(\wedge, =)$ -formula $G_1(x_0, \dots, x_{k-1})$ defining $G_1(\text{Pol}(M))$ we have:

$$Gp(x_0, \dots, x_{k-1}) \equiv G_1(x_0, \dots, x_{k-1}) \wedge \bigwedge_{0 \leq i < j \leq k-1} x_i \neq x_j,$$

that concludes the proof of the Lemma. $\square$

There is a slightly different way to obtain $G_1(\text{Pol}(M))$ by utilizing identities $\text{Pol}(M) = \text{Pol}(\mathcal{R}_M) = \bigcap_{1 \leq i \leq l} \text{Pol}(R_i)$ and $\text{Pol}(R_i) = \text{Pol}(R_i^*)$ ($i = 1, \dots, l$), where $\mathcal{R}_M = \{R_1, \dots, R_l\}$, and R_i^* is constructed by substituting R_i instead of $\bar{R}$ in (3.1). Hence, from the previous Lemma $G_1(\text{Pol}(R_i)) = \Delta_i R_i^*$ ($i = 1, \dots, l$), where Δ_i is an identification of variables in the formula defining R_i^* into k variables which is performed similarly to Δ . Finally, we obtain $G_1(\text{Pol}(M)) = \bigcap_{1 \leq i \leq l} G_1(\text{Pol}(R_i))$. Both procedures construct the same $(\wedge, =)$ -formula over $\mathcal{R}_M$ that defines $G_1(\text{Pol}(M))$. Note that the second procedure was presented in a compact form as a single formula in [1, Theorem 17].

Next, for an n -tuple $r = (r^1, \dots, r^n)$ ($n \geq 1$) over E we consider an n -ary relation $Gp(M, r) = \{(\varphi r^1, \dots, \varphi r^n) \mid \text{for all } \varphi \in \text{Aut}(M)\}$, that is, an n -ary relation which consists of images of r under all permutations from $\text{Aut}(M)$. Note that if $\text{Aut}(M)$ is a trivial unit group, then, clearly, $Gp(M, r) = \{r\}$.

Lemma 3.3. Let R be an n -ary ($n \geq 1$) relation from $\text{InvAut}(M)$ on a finite structure M . Then R can be defined by a positive existential formula constructed using $|R| \cdot O(n^2)$ steps from the k -orbit $G_p(M)$, which contains no more than $(k-1) \cdot |R|$ existential quantifiers.

Proof of the Lemma. First, for every n -tuple $r \in R$ we construct from $G_p(M)$ the n -ary relation $G_p(M, r)$ by a $(\exists, \wedge, =)$ -formula (p.p. formula) in such way that transforms k -tuple $(0, 1, \dots, k-1) \in G_p(M)$ into n -tuple r .

If r has all elements from E as its components, then $G_p(M, r)$ is defined by a $(\wedge, =)$ -formula applying to the formula $G_p(x_0, \dots, x_{k-1})$ that defines $G_p(M)$ a conjunction with the equality corresponding to equal coordinates in r (in case they exist). Then we perform a permutation of coordinates to align the resulting n -tuple obtained from $(0, \dots, k-1)$ with r . In case, when $\{r^1, \dots, r^n\} = E' \subseteq E$, we, at first, apply to the predicate $G_p(x_0, \dots, x_{k-1})$ a string of existential quantifiers with indices of variables covering the whole set E/E' . Hence, there are $O(n^2)$ steps, such as application of an existential quantifier, conjunction with equality and permutation of n coordinates to obtain r from a given n -tuple (requires at most $n(n-1)/2$ comparisons of pairs of elements from E and, consequently, interchanging corresponding coordinates in a relation), to construct the formula defining $G_p(M, r)$ from $G_p(M)$.

The following example may clarify this procedure. Let $E = \{0, 1, 2\}$ and $r = (2, 0, 0)$. Then we have $G_p(r)(x_2, x_1, x_0) \equiv \exists x G_p(x_0, x, x_2) \wedge x_0 = x_1$. Namely, we apply $\exists x G_p(x_0, x, x_2)$, since 1 is not a coordinate in r and stands at the second place in $(0, 1, 2)$, then we perform a conjunction with $x_0 = x_1$ (here, $(0, 1, 2)$ becomes $(0, 0, 2)$), along with permutation of variables $\Delta: \{x_0, x_1, x_2\} \rightarrow \{x_2, x_1, x_0\}$ to align $(0, 0, 2)$ with $(2, 0, 0)$.

Next, it is easy to prove that for an n -ary R the property $R \in \text{InvAut}(M)$ is equivalent to:

$$R = \bigcup_{r \in R} G_p(M, r), \quad (3.3)$$

which is defined from $G_p(M)$ by a positive existential n -ary formula, namely, as a finite disjunction of n -ary p.p. formulas over $G_p(M)$ (since operations $\exists$ and $\vee$ commute and we assume that each disjunctive term must have distinct bounded variables). Hence, we have the following positive existential formula over $\mathcal{R}$ and $x \neq y$ that defines R :

$$R(x_0, \dots, x_{n-1}) \equiv \bigvee_{r \in R} G_p(r)(x_0, \dots, x_{n-1}) \quad (3.4)$$

Clearly, the existential formula which defines R in (3.4) contains no more than $(k-1) \cdot |R|$ existential quantifiers. $\square$

Combining these two Lemmas we obtain the proof of the Theorem. $\square$

Recall that p.p. formula over $\mathcal{R} \cup \{x = y \vee x = z\} \cup \{x \neq y\}$ developed in Proposition 2.3 for every repetition-free relation $R \in \text{InvAut}(M)$ contains exactly $k^{|R|} - n$ existential quantifiers, that, clearly, is not a polynomial from $|R| \cdot n$.

Remark. Note that the number of steps to get G_p , that defines the relation $G_p(M)$, by a $(\wedge, =)$ -formula over the set $\{\mathcal{R}, \neq\}$ is an additive (see Theorem 3.1) to the number of steps to obtain a positive existential formula (3.4) over G_p that defines R , and, what is even more important, does not depend on the input R of the second, as well as the whole algorithm. So, it does not affect the tractability of this problem.

Note that the proof of this Theorem can be utilized in the case $R \in \text{InvEnd}(M)$, namely, we obtain a $(\wedge, =)$ -formula defining $G_1(\text{Pol}(M)) = G_1(\text{End}(M))$ from Lemma 3.2 (see (3.2)), then replace (3.3) by stipulation $R = \bigcup_{r \in R} G_1(M, r)$, where a k -ary relation $G_1(M, r) = \{(\psi r^1, \dots, \psi r^n) \mid \text{for all } \psi \in \text{End}(M)\}$ can be obtained as a p.p. definable relation from $G_1(\text{End}(M))$ following Lemma 3.3. Hence, we have the following Proposition.

Proposition 3.4. Let M be a finite structure of cardinality $k, k \geq 2$, having a finite signature. Then there exists an algorithm such that for every n -ary ($n \geq 1$) $R \in \text{InvEnd}(M)$, by using $|R| \cdot O(n^2)$ steps, it presents a positive existential formula, with no more than $(k-1) \cdot |R|$ existential quantifiers, by the set $\mathcal{R}$ of all unnested atomic predicates from M , which defines R .

Note that the upper estimate of the number of quantifiers in Theorem 3.1 and Proposition 3.4, that is $(k-1) \cdot |R|$, does not depend on the signature of a structure M , but only on its cardinality. Therefore, such estimate holds for any finite signature structure on the domain of finite cardinality k ($k \geq 2$).

Now we show that this is also true for M with an infinite signature.

Proposition 3.5. Let be a finite structure M of cardinality $k, k \geq 2$. Then the main k -orbit $G_p(M)$ of $\text{Aut}(M)$ is defined by a $(\wedge, =)$ -formula by the over $\mathcal{R}$ of all unnested atomic predicates over M and $x \neq y$, and, as a result, every n -ary ($n \geq 1$) relation R on the domain of M , that can be defined by the first order formula over M , is constructed by a positive existential formula over $\mathcal{R}$ and $x \neq y$ with no more than $(k-1)|R|$ existential quantifiers.

Proof. The case of a finite signature is in Lemmas 3.2 and 3.3. Then assume that $\mathcal{R}$ is an infinite set of predicates expressing distinct relations, which means that M has an infinite signature. Set $\mathcal{R}_M = \bigcup_{n \geq 2} \mathcal{R}_M^{(n)}$, where $\mathcal{R}_M^{(n)}$ are all m -ary ($1 \leq m \leq n$) unnested atomic relations. Clearly, $\text{Aut}(\mathcal{R}_M^{(n+1)}) \subseteq \text{Aut}(\mathcal{R}_M^{(n)})$ ($n \geq 2$). Then since all $\text{Aut}(\mathcal{R}_M^{(n)})$ are finite sets with non-void intersection the chain $\{\text{Aut}(\mathcal{R}_M^{(n)})\}$ stabilizes at some $n = n_0$. Hence, $\text{Aut}(\mathcal{R}_M^{(n_0)}) = \text{Aut}(\mathcal{R}_M)$ and applying Lemma 3.2 we obtain the relation $G_p(M)$ defined by a $(\wedge, =)$ -formula from $\mathcal{R}_M^{(n_0)}$ and $x \neq y$. Then we apply Lemma 3.3 $\square$

Then from (3.1), Lemma 3.3 and Proposition 3.5, as well as from $|R| \leq k^n - 1$ for a non-full n -ary relation R we obtain a statement on the boundary for a size of quantifier prefix in the first order formulas over finite structures (the case of full relations does not require quantifiers at all, rather a conjunction with the unary full relation $x = x$).

Proposition 3.6. *Let M be a finite structure of cardinality $k, k \geq 2$, having either a finite or an infinite signature. Then each n -ary ($n \geq 1$) first order formula over the atomic predicates from M is equivalent to a positive existential formula over $\{\mathcal{R}, x \neq y\}$ with the prefix of at most $(k - 1)(k^n - 1)$ existential quantifiers.*

Next, we use relations of the kind $G_p(M, r)$ ($r \in R$) to establish a cubic algorithm on decidability of $\text{InvAut}(M)$. First, we use three elementary operations on matrices of relations to construct every $G_p(M, r)$ ($r \in R$) from $G_p(M)$. Namely, (1) addition of an equal column to the matrix (corresponds to conjunction with the equality); (2) interchanging columns (permutation of coordinates in a relation); (3) crossing out a column (projection or, an application of an existential quantifier). Then from Lemma 3.2 we obtain that each $G_p(M, r)$ ($r \in R$) is constructed by $O(n^2)$ steps from $G_p(M)$. In total, $|R| O(n^2)$ steps are utilized for all $G_p(M, r)$ ($r \in R$).

Proposition 3.7. *A relation R on the domain of M has the property $R \in L_{\text{ow}}(M)$ if and only if $G_p(M, r) \subseteq R$ for every $r \in R$.*

Proof. Clearly, the inclusion $G_p(M, r) \subseteq R$ for every $r \in R$ is equivalent to $R = \bigcup_{r \in R} G_p(M, r)$, which, in turn, is equivalent to $R \in \text{InvAut}(M)$ (see Lemma 3.3). Then we apply Proposition 2.1(1). $\square$

Note that the inclusion $G_p(M, r_0) \subseteq R$ for some $r_0 \in R$ can be verified by comparing of no more than $|R|^2$ n -tuples, that is, by comparing of no more than $n|R|^2$ numbers. This follows from the fact that $|G_p(M, r_0)| > |R|$ for some $r_0 \in R$ immediately implies $R \notin L_{\text{ow}}(M)$. Hence, to check the inclusion property (if the previous condition does not hold) we consider only relations with $|G_p(M, r)| \leq |R|$. So, in total, to verify the above Proposition we need no more than $n|R|^3$ steps. We summarize the results of this Section.

Theorem 3.8. *Let M be a finite structure with a finite signature. Then there is a cubic algorithm such that for every n -ary ($n \geq 1$) relation R on the domain of M it recognizes (in $O(n^2|R|^3)$ steps) whether R can be defined by the first order predicate formula over M , and if it is true, creates (in $|R| O(n^2)$ steps) a positive existential formula over $\{\mathcal{R}, x \neq y\}$ with at most $(k-1)|R|$ existential quantifiers, which defines R .*

Similar result holds for the closure $\text{InvEnd}(M)$, but instead of a formula over $\mathcal{R}$ and $x \neq y$, we consider a positive existential formula over $\mathcal{R}$. In total, we may claim a cubic algorithm with the complexity $O(n^2|R|^3)$ (see the previous theorem).

Theorem 3.9. *Let M be a finite structure with a finite signature. Then there is a cubic algorithm, with the complexity $O(n^2|R|^3)$, such that for every n -ary ($n \geq 1$) relation on the domain of M it recognizes whether R can be defined by a positive existential formula in the signature of M , and if it is true, creates in $|R| O(n^2)$ steps a positive existential formula over $\mathcal{R}$ with no more than $(k-1)|R|$ existential quantifiers, which defines R .*

There is a question which naturally arises:

Question. Is it possible to lower the estimate $(k-1)|R|$ for the number of existential quantifiers in the presentation of n -ary ($n \geq 1$) relations from $L_{\text{ow}}(M)$ by the n -ary first order formulae, that is, by $(\exists, \wedge, \vee, \neg, =)$ -formulae over atomic predicates from M ?

Data availability

No data was used for the research described in the article.

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